

2023 AMC 10A Problem 10 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10A Problem 10. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10A Problem 10

**Problem:**

Maureen is keeping track of the mean of her quiz scores this semester. If Maureen scores an 11 on the next quiz, her mean will increase by 1. If she scores an 11 on each of the next three quizzes, her mean will increase by 2. What is the mean of her quiz scores currently?

Answer Choices:

- A. 4
- B. 5
- C. 6
- D. 7
- E. 8

Solution:

Suppose Maureen has taken n quizzes so far, with mean m . Then

$$\frac{mn + 11}{n + 1} = m + 1 \quad \text{and} \quad \frac{mn + 3 \cdot 11}{n + 3} = m + 2$$

These two equations simplify to $11 = m + n + 1$ and $33 = 3m + 2n + 6$, respectively. Subtracting twice the first equation from the second equation yields $11 = m + 4$, so $m = \text{(D)} \boxed{7}$. Then $n = 3$, so

Maureen has taken 3 quizzes so far, and their scores could, for example, be 6, 7, and 8. Then one score of 11 increases the mean to

$$\frac{6 + 7 + 8 + 11}{4} = \frac{32}{4} = 8$$

and three scores of 11 increase the mean to

$$\frac{6 + 7 + 8 + 11 + 11 + 11}{6} = \frac{54}{6} = 9$$

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